

Basic Descriptive Statistics

Topic Outline

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This document outlines some basic statistics topics from identifying types of data to exploring some techniques used to interpret the nature of a dataset or sample.

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1 Foundations: Population, Sample, and the Scope of Statistics

- **Population** — the complete, defined group relevant to a decision or study, not necessarily a literal town, state, or country population.
- **Sample** — a subset drawn from the population, used to make inferences about it.

Concept map. Statistics is divided into two branches:

- **Descriptive Statistics** — organizing, summarizing, and presenting data. Covers: types of data, data handling, visualization, measures of central tendency, measures of dispersion, measures of position, and correlation.
- **Inferential Statistics** — drawing conclusions about a population from sample data.

2 Types of Data

2.1 By Source

- **Primary data** — collected first-hand (surveys, observation, case studies).
- **Secondary data** — already collected by someone else and reused.

2.2 By Organization

- **Raw data** — unprocessed, as collected.
- **Organized / classified data** — arranged into a usable structure (e.g. a table).

2.3 By Values

- **Discrete data** — countable, distinct values.
- **Continuous data** — can take any value within a range.

2.4 By Number of Variables

- **Univariate** — one variable (e.g. the height of every student in a class).
- **Bivariate** — two variables studied together.
- **Multivariate** — more than two variables studied together.

Note

The unit illustrates these ideas with two applied activities: logging one's own weekly/monthly pocket expenditure, and reading the regional-coding logic built into an Indian postal PIN code.

3 Levels of Measurement

Four measurement scales are covered, in increasing order of information content.

3.1 Nominal Scale

Categorizes data by name or type only; categories are mutually exclusive and exhaustive, with no ranking implied (e.g. subject taught, gender, religion).

3.2 Ordinal Scale

Categories can be ranked, but the size of the gap between ranks is not precisely measurable (e.g. a 5-point mood scale, or a performance rating from “very poor” to “very good”).

3.3 Interval Scale

Adds equal, meaningful intervals between values, but has no true zero — values below zero are meaningful (e.g. coordinate positions on a Cartesian plane).

3.4 Ratio Scale

Has all interval-scale properties plus a true, meaningful zero; values never fall below zero (e.g. height, weight, area, length, time duration).

Scale	Adds	Suitable graphs	Central tendency	Dispersion
Nominal	Naming only	Bar graph, pie chart	Mode	Binomial/multinomial variance
Ordinal	+ order	Bar graph, histogram	Median	Range
Interval	+ equal spacing	Histogram (area-based)	Mean	Standard deviation
Ratio	+ true zero	Histogram (area-based)	Geometric/harmonic mean	Coefficient of variation

Table 1: *

Classification adapted from Afifi, May & Clark, *Practical Multivariate Analysis*, 5th ed. (2012).

4 Graphical Representation of Data

4.1 Bar Graph

Compares categories by bar height; suits a non-numerical (categorical) independent variable.

4.2 Pie Chart

A circle divided into sectors (“angular circle diagram”); used to compare each component’s share against the whole. Total values correspond to 360° of arc.

4.3 Line Graph

Connects successive data points to show how a variable changes, typically against time; multiple lines can share one set of axes for comparison.

4.4 Histogram

Consecutive bars with no gaps, used only for *continuous* class intervals — never for a discrete variable. Bar height reflects the class frequency.

Key Formula

For **unequal class intervals**, frequencies must be adjusted before plotting:

$$\text{Adjusted frequency} = \frac{\text{Minimum class size}}{\text{Class size}} \times \text{Frequency}$$

4.5 Frequency Polygon

Built by plotting class midpoints against frequency and joining them (often on top of a histogram, with one zero-frequency interval added at each end); used to visualize the overall shape of a distribution.

4.6 Choosing a Graph

- Bar graph — comparison across non-numerical categories.
- Pie chart — parts of a whole at one point in time (no trend over time).
- Line graph — continuous change over time.
- Histogram — distribution across continuous class intervals.
- Frequency polygon — shape of a distribution.

4.7 Charts in Excel

Highlight the data → **Insert** tab → choose the chart type.

5 Measures of Central Tendency

5.1 Mean

The arithmetic average; uses every value, so it is sensitive to outliers.

Key Formula

$$\bar{x} = \frac{\text{Sum of all observations}}{\text{Number of observations}} \quad \text{Excel: } =\text{AVERAGE}(\text{range})$$

For grouped data, the mean can be found by the *direct method*, the *assumed mean method*, or the *step-deviation method*.

5.2 Median

The middle value once data is ordered; a positional average, so it is unaffected by outliers.

- Odd n : median = $\left(\frac{n+1}{2}\right)^{\text{th}}$ observation.
- Even n : median = average of the $\left(\frac{n}{2}\right)^{\text{th}}$ and $\left(\frac{n}{2} + 1\right)^{\text{th}}$ observations.

Excel: =MEDIAN(range)

5.3 Mode

The most frequently occurring value. A data set can be *uni-modal* (one mode), *bi-/multi-modal* (two or more values tied for most frequent), or have *no mode* (every value equally frequent).
Excel: =MODE(range)

Note

For a moderately skewed, uni-modal distribution: Mean – Mode = 3(Mean – Median).

6 Measures of Dispersion

Dispersion describes how spread out the data is around its central value.

6.1 Range

The simplest measure: the gap between the largest and smallest observations.

6.2 Quartile Deviation

Key Formula

$$Q.D. = \frac{Q_3 - Q_1}{2}, \quad \text{Interquartile Range (IQR)} = Q_3 - Q_1$$

The IQR is the “middle half” of the data and is used to flag outliers: any value more than $1.5 \times \text{IQR}$ above Q_3 or below Q_1 is an outlier. Excel: $Q_1 = \text{PERCENTILE}(\text{range}, 0.25)$, $Q_3 = \text{PERCENTILE}(\text{range}, 0.75)$.

6.3 Mean Deviation

Key Formula

$$\text{M.D.}(\bar{x}) = \frac{1}{N} \sum_{i=1}^n |x_i - \bar{x}|, \quad \text{M.D.}(\text{median}) = \frac{1}{N} \sum_{i=1}^n |x_i - M|$$

Defined for both ungrouped and grouped (discrete or continuous) data, measured either about the mean or about the median.

6.4 Variance

Key Formula

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2$$

The arithmetic mean of squared deviations from the mean — unlike range, MD, or QD, it combines every value into a single measure of spread.

6.5 Standard Deviation

Key Formula

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2} \quad \text{Excel: } =\text{STDEV}(\text{range})$$

Low SD $\Rightarrow$ values cluster tightly around the mean (more reliable); high SD $\Rightarrow$ values are widely spread. SD is always positive and is sensitive to outliers.

7 Skewness

Skewness measures the asymmetry of a distribution.

- **Positive skew** — longer tail on the right.
- **Negative skew** — longer tail on the left.
- **Zero skew** — symmetrical distribution.

Key Formula

Coefficients of skewness:

$$\text{Pearson's 1st} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$$

$$\text{Pearson's 2nd} = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$$

$$\text{Quartile} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$\text{Percentile} = \frac{P_{90} + P_{10} - 2 \text{Median}}{P_{90} - P_{10}}$$

$$\text{Decile} = \frac{D_9 + D_1 - 2 \text{Median}}{D_9 - D_1}$$

8 Kurtosis

Kurtosis describes the “peakedness” of the frequency curve. Excel: =KURT(range).

- **Leptokurtic** (positive) — a sharp, heavy-tailed peak.
- **Platykurtic** (negative) — a blunt, light-tailed peak.
- **Mesokurtic** (zero) — a moderate peak, matching the normal distribution.

9 Measures of Position

9.1 Percentile Rank

The P^{th} percentile is the value below which $P\%$ of observations fall. Percentiles $P_1, \dots, P_{99}$ divide the data into 100 equal groups.

9.2 Quartiles

- Q_1 — up to 25% of data; Q_2 — 25–50% (up to the median); Q_3 — 50–75%; Q_4 — 75–100%.
- Equivalences: 25th percentile = Q_1 ; 50th percentile = median = Q_2 ; 75th percentile = Q_3 .

Key Formula

Ungrouped (individual series):

$$Q_1 = \frac{1}{4}(n+1)^{\text{th}} \text{ value}, \quad Q_3 = \frac{3}{4}(n+1)^{\text{th}} \text{ value}$$

Grouped data:

$$Q_1 = l + \left(\frac{\frac{n}{4} - cf}{f} \right) \times h, \quad Q_3 = l + \left(\frac{\frac{3n}{4} - cf}{f} \right) \times h$$

where l = lower class boundary, cf = cumulative frequency before the class, f = class frequency, h = class width.

10 Correlation

Correlation measures the strength and direction of a linear relationship between two variables via the correlation coefficient r .

- $r = -1$: perfect negative correlation. $r = 0$: no correlation. $r = +1$: perfect positive correlation.
- $r > 0$: positive/direct relationship. $r < 0$: negative/inverse relationship.

Excel: =CORREL(range1, range2)

Key Formula

Karl Pearson's coefficient of correlation:

$$\text{Direct (ungrouped): } r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2}}$$

$$\text{Direct (grouped): } r = \frac{N \sum XY - \sum X \sum Y}{\sqrt{N \sum X^2 - (\sum X)^2} \cdot \sqrt{N \sum Y^2 - (\sum Y)^2}}$$

$$\text{Assumed mean method: } r = \frac{N \sum d_x d_y - \sum d_x \sum d_y}{\sqrt{N \sum d_x^2 - (\sum d_x)^2} \cdot \sqrt{N \sum d_y^2 - (\sum d_y)^2}}$$

11 Quick-Reference Summary

Measure	Definition	Symbol
Mean	Sum of values $\div$ number of values	$\bar{x}$
Median	Middle point of an ordered data set	M
Mode	Most frequent data value	—

Table 2: *

Mean is the most commonly used measure of central location.

Graph types covered: bar graph, pie chart, line graph, histogram, frequency polygon.

Note

Practical tool: Excel's *Data Analysis ToolPak* (enabled via File $\rightarrow$ Options $\rightarrow$ Add-ins) provides a one-click *Descriptive Statistics* report — mean, standard deviation, and related summary statistics — for a selected data range.